

Bisal Prasad

bisalprasad2003@gmail.com | +91 9395360772 | [LinkedIn](#) | [GitHub](#) | [LeetCode](#)

EDUCATION

National Institute of Technology, Silchar

B.Tech in Electronics and Communication Engineering

2023 – 2027

CGPA: 9.0/10

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, Verilog, SQL

Frameworks/Web: React.js, Next.js, Node.js, Express.js, Redux, MongoDB, Tailwind

Core Concepts: Data Structures & Algorithms, OOPS, OS (CPU Scheduling), DBMS, System Design

Developer Tools: Git, Docker, AWS (S3, Rekognition), Google Cloud (Vertex AI), VS Code

KEY SOFTWARE PROJECTS

CPU Scheduling Simulator | C++, OS Concepts [\[GitHub\]](#) | [\[Live Demo\]](#)

Oct 2025

- Engineered a high-performance simulator in C++ implementing **7** core scheduling algorithms (FCFS, RR, MLFQ) capable of handling **50+** concurrent processes.
- Developed a dual-mode interface (CLI & Web/Next.js) visualizing PCB states, reducing concept grasp time by **40%**.
- Optimized wait-time calculation logic using STL Priority Queues, achieving $O(n \log n)$ efficiency and **<10ms** latency.

LUXORA - E-Commerce Platform | MERN, Redux [\[GitHub\]](#) | [\[Live Demo\]](#)

Nov 2025

- Built a secure platform with **JWT Auth** and Role-Based Access Control, supporting **100+** distinct product SKUs.
- Implemented global state management via **Redux Toolkit**, reducing redundant API calls by **30%** for cart operations.
- Designed scalable REST APIs in Node.js/Express, achieving sub-second load times (**<500ms**) for real-time reviews.

Amazon GreenX | AWS Rekognition, ML, CV [\[GitHub\]](#) | [\[Live Demo\]](#)

Jun 2025

- Integrated **Machine Learning** pipelines using **AWS Rekognition** to analyze **500+** images for carbon footprint scoring with **95%** accuracy.
- Created a 3D CV bin-packing optimizer, reducing packaging waste by **35%**; selected as **Top 130** in Amazon HackOn.

EXPERIENCE

Research Intern - BCI Signal Processing [\[GitHub\]](#)

May 2025 – Jul 2025

National Institute of Technology, Silchar

On-site

- Processed **122-channel** EEG data using Wavelet Transforms; built a Deep Learning model with **90%+** accuracy.
- Engineered a 'Thought-to-Speech' pipeline, achieving **2x** faster neural decoding for assistive communication.

Machine Learning Intern

Jun 2024 – Jul 2024

Academor Edtech

Remote

- Improved Loan Prediction model accuracy by **15%** via hyperparameter tuning on a dataset of **10k+** records.
- Deployed an NLP sentiment analysis pipeline, achieving **98%** precision on customer feedback data.

ACHIEVEMENTS & CODING PROFILE

Coding Profile: [LeetCode](#) (300+ Solved), [CodeForces](#) (Pupil), GeeksforGeeks (50+).

Winner, Neurathon 2025: Built 'CALL.E', an AI calling agent using Llama-3, reducing latency by **45%**.

Finalist, Google Cloud Agentic AI Day: Top 50 in India. Built 'Project Kisan' (Vertex AI) for 1M+ farmers.

Winner, Drone Innovation Challenge: Prototyped a GPS-independent drone with **99.9%** signal reliability.

LEADERSHIP

General Secretary

Apr 2025 – Present

IEEE Robotics and Automation Society, NIT Silchar

- Led a **50-member** team to organize **5+** technical workshops, impacting **500+** students across campus.
- Expanded a GPS-independent fire-surveillance drone to enhance emergency response capabilities.